Define polynomials $\pi(x) = \sum_{j=0}^{\mathrm{N_c}-1} \alpha_j x^j$, $\sigma(x) = \sum_{j=0}^{\mathrm{N_c}-1} \beta_j x^j$, then, for each j, we have

$$\pi(\lambda_j)\hat{q}_j + \lambda_j\sigma(\lambda_j)\hat{p}_j = 0, \qquad \lambda_j\pi(\lambda_j)\hat{p}_j - \lambda_j\sigma(\lambda_j)\hat{q}_j = 0,$$

multiplying the first equation by $\hat{q}_j$ and the second by $\hat{p}_j$ and adding, we obtain

$$[\hat{q}_j^2 + \lambda_j\hat{p}_j^2]\pi(\lambda_j) = 0.$$

The first factor is nonzero by the condition of being in $\mathcal{D}$ and the positive definiteness of $\boldsymbol{A}$, thus we must have $\pi(\lambda_j) = 0$ for all $\mathrm{N_c}$ distinct λ_j. But as π has degree $\mathrm{N_c} - 1$, this is a contradiction. Thus the collection must be linearly independent.

Demonstrating the Hörmander condition is sufficient for Assumption 1(ii) in Sect. 6.4.4. Assumption 1(i) can be verified by proving that solutions are able to access an open set around any point $\boldsymbol{y} \in \mathcal{D} \times \mathbb{R}$. We omit the proof of Assumption 1(i) for Nosé-Hoover-Langevin dynamics as it is follows the same structure as the proof of Lemma 6.1 for Langevin dynamics. The details of the proof can be found in [283], with examples for more general SDEs given in [256, 257].

The final requirement for geometric ergodicity (specified in Assumption 2 in Sect. 6.4.4) requires a function $\phi : \mathcal{D} \times \mathbb{R} \to (0, \infty)$ such that

$$\mathcal{L}\phi \leq -\alpha\phi + \delta,$$

for positive constants α and δ and where $\mathcal{L}$ is the generator of the stochastic process. Such a function is called a Lyapunov function.

The generator for Nosé-Hoover-Langevin dynamics (8.27)–(8.29), acting on a suitable test function $f(\boldsymbol{q}, \boldsymbol{p}, \xi)$ is

$$\mathcal{L}_{\mathrm{NHL}} f = \boldsymbol{M}^{-1}\boldsymbol{p} \cdot \nabla_q f - \nabla U(\boldsymbol{q}) \cdot \nabla_p f - \xi \boldsymbol{p} \cdot \nabla_p + \mu^{-1}\left[\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d}\mathrm{k_B}\mathrm{T}\right]\frac{\partial f}{\partial \xi}$$
$$- \gamma\xi\frac{\partial f}{\partial \xi} + \mathrm{k_B}\mathrm{T}\gamma\mu^{-1}\frac{\partial^2 f}{\partial \xi^2}.$$

If we consider $\boldsymbol{M} = \boldsymbol{I}$ for simplicity, a Lyapunov function is given in [283] as

$$\phi(\boldsymbol{q}, \boldsymbol{p}, \xi) = \frac{1}{2}\|\boldsymbol{p}\|^2 + U(\boldsymbol{q}) + \frac{\mu}{2}(\xi - a)^2,$$

where a is a constant. As our potential is smooth, it is reasonable to make the assumption that

$$1 < U_{\min} \leq U(\boldsymbol{q}) \leq U_{\max} < \infty, \quad \forall \boldsymbol{q} \in \mathcal{Q},$$

for some constants $U_{\min}$ and $U_{\max}$. Hence ϕ is strictly positive over the entire space, and we can compute that

$$\begin{aligned}\mathcal{L}_{\mathrm{NHL}}\phi &= \boldsymbol{p}\cdot\nabla U(\boldsymbol{q}) - \nabla U(\boldsymbol{q})\cdot\boldsymbol{p} - \xi\|\boldsymbol{p}\|^2 + \left[\|\boldsymbol{p}\|^2 - \mathrm{N_d k_B T}\right](\xi - a)\\ &\quad - \gamma\mu\xi(\xi - a) + \mathrm{k_B T}\gamma,\\ &\leq -a\|\boldsymbol{p}\|^2 - \gamma\mu\xi^2 + (a\gamma\mu - \mathrm{N_d k_B T})\,\xi + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma.\end{aligned}$$

Choosing a so that $a\gamma\mu - \mathrm{N_d k_B T} = 0$, we obtain

$$\mathcal{L}_{\mathrm{NHL}}\phi = -a\|\boldsymbol{p}\|^2 - \gamma\mu\xi^2 - \gamma\mu a^2 + \gamma\mu a^2 + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma.$$

Using $-\xi^2 - a^2 \leq -(\xi - a)^2/2$ gives

$$\mathcal{L}_{\mathrm{NHL}}\phi \leq -2a\frac{1}{2}\|\boldsymbol{p}\|^2 - \gamma\mu\frac{1}{2}(\xi - a)^2 + \gamma\mu a^2 + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma.$$

Now choosing

$$\alpha = \min\left(2a, \gamma\mu\right),$$

we have

$$\begin{aligned}\mathcal{L}_{\mathrm{NHL}}\phi &\leq -\alpha\phi + \alpha U(\boldsymbol{q}) + \gamma\mu a^2 + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma,\\ &\leq -\alpha\phi + \delta,\end{aligned}$$

for constants α and δ as required, where

$$\delta = \alpha U_{\max} + \gamma\mu a^2 + a\mathrm{N_d k_B T} + \mathrm{k_B T}\gamma.$$

This gives the necessary Lyapunov condition for the Nosé-Hoover-Langevin equations, and hence the dynamics are geometrically ergodic. Solution trajectories will sample the extended canonical distribution, and it remains to compute solutions using an appropriate discretization scheme.

8.4.2 Numerical Integration of the Nosé-Hoover-Langevin Equations

Many older schemes for Nosé-Hoover and Langevin dynamics make use of computational shortcuts to (for example) reduce the number of computed random values per iteration, reduce the data storage needed for an algorithm or replace functions that are costly to evaluate (e.g. exponential or trigonometric functions)